

De Morgan's laws

② $\Rightarrow$:

$$\forall x \in (E \cap F) \Rightarrow x \in E \text{ and } x \in F$$

$$x \in E \Rightarrow x \in F$$

$$x \in G \Rightarrow x \in F$$

$$x \in E \Rightarrow x \in E$$

$$\Leftarrow: \forall x \in E(F \cap G)$$

$$\Rightarrow x \in E \text{ and } x \in F \cap G$$

$$x \in F \Rightarrow x \in F$$

$$x \in G \Rightarrow x \in F$$

$$x \in F \Rightarrow x \in G$$

$\therefore$ left = right

1.4 distributive laws

$$① (E \cap F) \cap G = E \cap (F \cap G)$$

$$② (E \cap F) \cup G = (E \cup G) \cap (F \cup G)$$

i) if $x \in G \Rightarrow x \in E \cup (F \cap G) = \text{right}$

$$(\because E, F, G \subseteq E \cup (F \cap G))$$

1.2 commutative laws

ii) if $x \in E \cup F \Rightarrow x \in E$ or $x \in F$

$$\Rightarrow x \in E(F \cap G)$$

$$\text{if } x \in E \Rightarrow x \in \text{right}$$

$$\text{if } x \in F \Rightarrow x \in \text{right}$$

$\therefore x \in \text{right}$

$\Leftarrow$:

$$(\text{note } E, F, G \subseteq (E \cup F) \cap G = \text{left})$$

$$\forall x \in E \cup (F \cap G)$$

$$\Rightarrow x \in E \text{ or } x \in F \cap G$$

i) if $x \in E \Rightarrow x \in \text{left}$

ii) if $x \in F \cap G \Rightarrow x \in F$ or $x \in G$

$$\text{if } x \in F \Rightarrow x \in \text{left}$$

$$\text{if } x \in G \Rightarrow x \in \text{left} \text{ proof:}$$

$\therefore x \in \text{left}$

$\therefore$ left = right.

$$① E \cup F = F \cup E, E \cap F = F \cap E$$

proof:

$$① \forall x \in E \cup F$$

$$\Leftrightarrow x \in \text{at least one of them}$$

$$\Leftrightarrow x \in F \cup E$$

$$② \forall x \in E \cap F$$

$$\Leftrightarrow x \in \text{both of them}$$

$$\Leftrightarrow x \in F \cap E$$

1.3 associative laws

$$① (E \cup F) \cup G = E \cup (F \cup G)$$

$$② (E \cap F) \cap G = E \cap (F \cap G)$$

$$\Rightarrow \forall x \in (E \cup F) \cup G$$

$$\Rightarrow x \in E \cup F \text{ or } x \in G$$

2. De Morgan's laws

$$\textcircled{1} \left(\bigcup_{i=1}^n E_i \right)^c = \bigcap_{i=1}^n E_i^c$$

$$\textcircled{2} \left(\bigcap_{i=1}^n E_i \right)^c = \bigcup_{i=1}^n E_i^c$$

proof:

$$\textcircled{1} \quad \forall x \in \left(\bigcup_{i=1}^n E_i \right)^c$$

$$\Leftrightarrow x \notin \bigcup_{i=1}^n E_i$$

$$\Leftrightarrow x \notin E_i, \forall i$$

$$\Leftrightarrow x \in E_i^c, \forall i$$

$$\Leftrightarrow x \in \bigcap_{i=1}^n E_i^c$$

$$\textcircled{2} \quad \because \left(\bigcup_{i=1}^n E_i^c \right)^c = \bigcap_{i=1}^n (E_i^c)^c$$

$$= \bigcap_{i=1}^n E_i$$

$$\therefore \left(\bigcap_{i=1}^n E_i \right)^c = \bigcup_{i=1}^n E_i^c$$

$\textcircled{2} \Rightarrow$:

$$\forall x \in (EF) \cup G = \text{left}$$

$$\Rightarrow x \in EF \text{ or } x \in G$$

$$\Rightarrow \begin{cases} x \in E \text{ or } x \in G \Leftrightarrow x \in E \cup G \\ x \in F \text{ or } x \in G \Leftrightarrow x \in F \cup G \end{cases}$$

$$\Rightarrow x \in (E \cup G)(F \cup G) = \text{right}$$

$$\Leftrightarrow x \in (E \cup G)(F \cup G) = \text{right}$$

$$\Leftrightarrow \forall x \in (E \cup G)(F \cup G) = \text{right}$$

$$\Rightarrow \begin{cases} x \in E \cup G \Rightarrow x \in E \text{ or } x \in G \Leftrightarrow \\ x \in F \cup G \Rightarrow x \in F \text{ or } x \in G \end{cases}$$

$$\Rightarrow x \in E \text{ or } x \in F$$

$$\text{i) if } x \in G \Rightarrow x \in (EF) \cup G = \text{left} \quad \text{i) if } x \in E \Rightarrow x \in E, x \in G$$

$$\text{ii) if } x \notin G \Rightarrow x \in E, x \in F \Rightarrow x \in EF, x \in G$$

$$\Rightarrow x \in EF \Rightarrow x \in (EF) \cup G = \text{left} \Rightarrow x \in \text{left}$$

$\therefore \text{left} = \text{right}$.

$$\text{ii) if } x \in F \Rightarrow x \in F, x \in G$$

$$\Rightarrow x \in EF, x \in G$$

$$\Rightarrow x \in \text{left}$$

$\therefore \text{left} = \text{right}$.

proof: $\textcircled{1} \Rightarrow$:

$$\forall x \in (E \cup F) \cap G = \text{left}$$

$$\Rightarrow \begin{cases} x \in E \cup F \Rightarrow x \in E \text{ or } x \in F \\ x \in G \end{cases}$$

$$\text{i) if } x \in E, \forall x \in G, \therefore x \in E \cap G \Rightarrow x \in \text{right}$$

$$(\because E \cap G \subset E \cup F \cap G)$$

$$\text{ii) if } x \in F, \forall x \in G, \therefore x \in F \cap G \Rightarrow x \in \text{right}$$

$$\Leftrightarrow \text{if } x \in E \cup F \cap G = \text{right}$$

$$\Rightarrow x \in E \text{ or } x \in F \cap G$$